Derived Intersection Theory on Foliation

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In this paper, I'll discuss foliation from point of view of derived algebraic geometry. Derived algebraic geometry proves the ∞-equivalence of dg Lie algebroid and foliation. Now the quotient is defined by quotient of dg L_∞ groupoid generated from dg Lie albebroid, and it's calculated by dg L_∞ version of Chevalley-Eilenberg complex. I'm also wondering evaluation of the foliation using language of DT/GW invariants. First of all, the idea of foliation is classically defined differential geometrically on a smooth manifold:

Definition 0.1. Foliation

Foliation is a partition of a smooth manifold M into its submanifolds. One idea of generating foliation is to retrieve from distribution $D_x \subset TM$ for some $x \in M$, and this distribution classfies the points $x \in M$ to define foliaton. Distribution D_x is defined by its vector field $X, Y \in \Gamma(D_x)$ satisfying $[X, Y] \in \Gamma(D_x)$, and in this case D_x is called integrable. This D_x corresponds to a submanifold L, since there's its restriction $D_x|_L = T_x L$ by the unique maximal submanifold $L \subset M$ s.t. $x \in L$. Note in fact $D_x|_{L_y} = T_y L$ for all $y \in L$. Accordingly, the resulting set $\{L_y\}$ for $y \in M$ is the set of submanifolds $L_y \subset M$ and disjoint union $\coprod L_y = M$.

To verbalize the notion of foliation is an equivalent notion with integrability by Frobenius theorem, where integrability is done by Lie algebra. If Lie algebra describes the global property (there's no "if" in math Toynbee says), the corresponding Lie group quotient of manifold $[M/G]$ is a stack. It doesn't work, however. I'll use Lie algebroid to describe every point of the submanifold $L \subset M$, whose Lie groupoid $\mathcal{G}$ quotient is a quotient stack, which describes foliation $[M/\mathcal{G}]$. The first step of foliation my plan is to modernize the definition of foliation by using quotient stack with Lie algebroid/groupoid.

Definition 0.2. Groupoid

A groupoid is a category in which every morphisms are invertible. Denote the objects are $Ob(\mathcal{C}) = G_0$, and the morphisms are $Mor(\mathcal{C}) = G_1$, the double arrow $G_1 \rightrightarrows G_0$. For an arrow $g : x \to y$, there are source and target maps $s, t, : G_1 \to G_0$, together with composition $G_1 \times_{G_0} G_1 \to G_1$, identity arrow and inverses. A group becomes a Lie groupoid if G_0 and G_1 are smoooth manifolds, and all the maps are smooth.

There are many important examples of Lie groupoid. The trivials things (logically equivalent to Lie group but somehow interesting generalization) are Lie group, which is a Lie groupoid with one object so $G \rightrightarrows *$. Another example which is more geometric is that Lie group action $G \times X \rightrightarrows X$ can be modeled by $G_1 = G \times X$ and $G_0 = X$. By the context of foliation, Lie groupoid $Hol(\mathcal{F})$ of the foliation $\mathcal{F}$ is holonomy groupoid. All the above examples are important. Now we consider quotient stack by Lie groupoid, in particular by Lie group. The definition of quotient stack below is a moduli stack while keeping the group symmety structure for group G, but groupoid quotient $\mathcal{G}$ is somewhat more complicated:

Definition 0.3. Quotient Stack

For an algebraic variety X, quotient stack by Lie group G is $[X/G]$ is a functor $[X/G] : (Spaces)^{op} \to Grouoids$.

$$[X/G](S) = \{(P \to S, f : P \to X) | P \to S \text{ is a principal G-bundle}, fisG - equivariant\}$$

In particular if $X = pt$, the quotient stack is $[pt/G] = BG$, and $BG(pt)$ is one-object groupoid whose automorphism group is $G = Hom(*, *)$.

Meanwhile, quotient stack by groupoid $[U/R]$ for a groupoid R is a stackification $[U/R] = St([U/_pR])$ of category fibered in groupoids $p : U \to R$. This works for all $c \in R$, there exists $x \in U$ s.t. $p(x) = c$, and for all $f : x \to y$, $p(f) = id_c$.

where I abbreviate what the stackification means in this introductory chapter for the moment. In any case, the important quotient stack is holonomy groupoid quotient $[M/Hol(F)]$, which is a foliation. But how does Lie groupoid appears, where Lie algebra solves integrability condition in manifold? Lie algebra can be generalied to Lie algebroid by its globalization, where Lie groupoid comes. The definition of Lie algebroid is:

Definition 0.4. Lie Algebroid

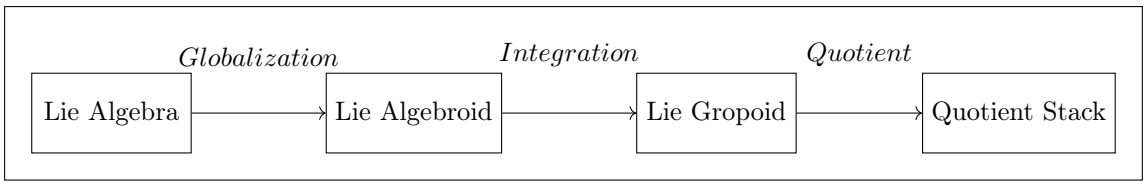
For a smooth manifold M, a Lie algebroid over M consists of $(A \to M, [\cdot,\cdot]_A, \rho)$ where $A \to M$ is a vector bundle, $[\cdot,\cdot]_A : \Gamma(A) \times \Gamma(A) \to \Gamma(A)$ is a Lie bracket of sections, $\rho : A \to TM$ is an anchor map. These satisfy product rule $[X, fY]_A = f[X, Y]_A + \rho(X)(f)Y$, and the anchor preserves bracket $\rho([X, Y]_A) = [\rho(X), \rho(Y)]$.

This Lie algebroid is a generalization of Lie algebra. In fact, the fiber A_x $x \in M$ of vector bundle $A \to M$ is Lie algebra, so containing the data of local symmetry.

Remark 0.1

As a comparison, quotient stack $[M/G]$ of gauge theory is a Lie group quotient, not Lie groupoid, where the local symmetry of particle action is everywhere the same, but the construction is nothing to do with the foliation problem. In other words, not all quotient stack is foliation.
The computation of quotient stack $[M/G]$ is done by Chevalley-Eilenberg complex. There's generalization of Chevalley-Eilenberg for the case of quotient stack by Lie groupoid.

In summary, the above argument can be depicted in the following flow chart, and now that the foliation is modernized geometrically, studying foliation is somewhat equivalent to studying quotient stack:



Several questions appear from this informal writing of foliation. First of all, to make the argument more precisely, can Lie algebroid be always integrated to the Lie groupoid? Now true in general, but Crainic-Fernandes integrability criterion claims Lie algebroid used in foliation can be always integrated.

Proposition 0.1. Crainic-Fernandes Criterion

TBD

Let's consider foliation algebraically. From Toën, derived foliation is

Definition 0.5. Affine Derived Foliation

TBD

The problem here is affine derived foliation works by complex dimension not in real, and this might be inconvenient for Lagrangian foliation for instance.

Definition 0.6. Affine Derived Foliation –Fixed

TBD

and the general derived foliation $Fol(X)$ is projective limiti of affine derived foliation $Fol(X) = \varprojlim Fol(Spec(A))$.

Question 0.1

Is there any connection of quotient stack by Lie groupoid and derived foliation, as geometry and algebra correspondence?

Consider the intersetion theory problem of quotient stack of the foliation. Intersection theory is to compare two different geometry on the same domain. Consider two quotient projection maps $p : X \to X/\mathcal{F}$ and $q : X \to X/\mathcal{G}$, which are foliations, and their algebra are contravariant $p^* : \mathcal{O}_{X/\mathcal{F}} \to \mathcal{O}_X$ and $q^* : \mathcal{O}_{X/\mathcal{G}} \to \mathcal{O}_X$, and it makes a pullback diagram below to make a fiber product $\mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}}$, that's how we define as the derived intersection:

$$\begin{array}{ccc} \mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}} & \longrightarrow & \mathcal{O}_{X/\mathcal{F}} \\ \downarrow & & \downarrow{\scriptstyle p^*} \\ \mathcal{O}_{X/\mathcal{F}} & \xrightarrow{q^*} & \mathcal{O}_X \end{array}$$

Now let's set some research questions. To understand its structure, I consider beginning with specific geometry. Subjectively, groupoid foliation is too broad, think of Lie group quotient in gauge theory as a comparison, Lie groupoid quotient is much more dirty than that of Lie group, so making the shape M as simple as possible, for instance, to make it clear, and possibly M could be just projective space $\mathbb{P}^n$ or a plane $\mathbb{A}^n$ or a hyperkähler manifold, I'd also study its computation using Chevalley-Eilenberg, Donaldson-Thomas/Gromov-Witten invariants or any other, while it's interesting tracking what's the missing data when we define quotient stack of Lie groupoid. On the other hand, we could also consider the categorification to make it more abstract. Typically, I noticed derived algebraic geometry might be useful to discuss these problems, and I'll consider geometrization of moduli stack of foliations where we can discuss intersection or any others:

Question 0.2

- Making geometry M as simple as possible?
- Computation of groupoid version of CE cohomology? DT/GW invariants?
- Does quotient stack contain all the necessary information?
- Is there either deformation or quantization or derived geometry?
- Categorification?

Another view point is that geometry is useful in QFT, for example. Especially, consider Lagrangian intersection theory of Calabi-Yau geometry, whose quantum cohomology generates Gromov-Witten invariant, and this

helps to compute path integral in both 4d/10d gauge theory. For this time I'll introduce derived Lagrangian intersection theory to this foliation. Suppose there are two distinct Lagrangian folations $\mathcal{F}_1$ and $\mathcal{F}_2$ on the same geometry M, and if considered intersection of these foliations, this might be interesting and different.

Furthermore, if the whole argument contains dgL_∞ algebroid as generalization of Lie algebroid, I wonder if there is any comparison with A_∞ side in particular Fukaya category, or if there's any connection to homological mirror symmetry? I'll simply summarize in the list:

Question 0.3

- What happens if the foliation is Lagrangian?
- Is there any connection to Fukaya category?

I'll skip the technical detail here, but to conjecturally answer these questions, I'll begin with Symplectic vector space and its generalization to derived algebraic geometry. Consider(informal) Lagrangian quotient in place of Lie groupoid quotient. Let Symplectic vector space be V with its 2-form ω and some Lagrangian $L \subset V$ s.t. $\omega|_L = 0$, the quotient V/L as a vector space is the baby form of Lagrangian foliation problem, where we have further generalization:

Definition 0.7

- step 1 (vector space):
  The quotient V/L is no longer a symplectic space, but it induces the pair $L \times V/L \to \mathbb{R}$ in addition the duality $L \cong (V/L)^*$ make it $(V/L)^* \times V/L \to \mathbb{R}$.
- step 2 (cotangent bundle):
  For a smooth manifold X, cotangent bundle T^*X is a symplectic manifold.
- step 3 (symplectic manifold):
  For the underived case, let a symplectic manifold X and its tangent bundle TX and Lagrangian L, the quotient TX/L replaces (V/L).
- step 4 (derived method):
  Globally, we use L_∞ groupoid quotient stack $[X/\mathcal{G}]$, and derived tangent complex $\mathbb{T}_{[X/\mathcal{G}]} \cong [\mathfrak{g}f \to TX]$ or noted by $\mathbb{T}_{\mathcal{F}}$ of derived foliation. the duality pairing will be $\mathbb{T}_{\mathcal{F}}^\vee \otimes \mathbb{T}_{\mathcal{F}} \to \mathcal{O}L_X$.

So far we defined Symplectic calculus using Lagrangian quotient, which works fine on vector space, but it doesn't work in general, since not all leaves on the Lagrangian foliation are the same. But is it still possible to trivialize Lagrangian foliation, so each leaves of that foliation is isomorphic to one fixed foliation up to deformation or any other? Locally yes, but it doesn't work in general for obstruction of holonomy and topology. However, supposely we assume very naïvely an identification of the quotient stack of foliation and some derived cotangent analogue in this formula [1], cotangent bundle has particularly important of general symplectic manifold. Below is a conjecture:

Conjecture 0.1. Symplectic Cotangent

$$[X/\mathcal{G}] \cong [T^*Y]_{der}$$

Now to introduce Nadler-Zaslow as an application of cotangent bundle T^*X, which is an analogue of HMS for the case of $Fuk(T^*X)$ using microlocal analysis.

Proposition 0.2. Nadler-Zaslow

There is an equivalence of category

[1] prove it

$$Fuk(T^*X) \cong Sh_c(X)$$

where c means constructible not compactness.

I'd also put generalization [2] to derived algebraic geometry.

Conjecture 0.2. Nadler-Zaslow Enhanced

$$Fuk(\mathbb{T}_X) \cong Sh_c(X)$$

For the organization of this paper, I begin with language of derived algebraic geometry and ∞ categories, then enhancing groupoid foliation in derived language as well as classically, and I add Lagrangian intersection theory.

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1 Derived Foliation

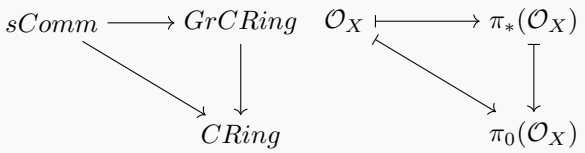
Definition 1.1. Graded Mixed CDGA

For defining graded mixed cdga, cdga is a graded commutative $xy = (-1)^{|x||y|}yx$ and dg algebra, and let graded mixed object E be a $\mathbb{Z}$-graded object $E = \oplus E(n)$ with extra differential $\epsilon_n : E(n) \to E(n+1)[-1]$ or $\epsilon : E \to E((1))[-1]$ s.t. $\epsilon^2 = 0$. Each $E(n)$ has a differential $d : E(n)^i \to E(n)^{i+1}$. ∞-category of graded mixed objects are denoted by $\epsilon - dg^{gr}$, and if E is also cdga, it forms category $\epsilon - cdga^{gr}$. I'll denote category of derived affine scheme $dAff^{\mathcal{F}} = \epsilon - cdga^{gr}$ as contravariant of that of the graded mixed cdga. In addition, consider an ∞-functor $DR : cdga \to \epsilon - cdga^{gr}$ where $DR(A) = Sym^u_{A'}(\Omega^1_{A'}[1])$. This DR is functorial in the sense of ∞-categories, and analogously, $DR(X) = Sym_{\mathcal{O}_X}(\mathbb{L}_X[-1])$.

Definition 1.2. Derived Scheme

A derived scheme $(X, \mathcal{O}_X)$ consists of functor $\mathcal{O}_X : Open(X)^{op} \to sComm$, mapping $U \mapsto \mathcal{O}_X(U)$ where $sComm$ is an ∞-category of simplicial commutative rings (also called derived rings) defined by $sComm := Fun(\Delta^{op}, CRing)$. This $sComm$ can be realized as a category of graded commutative ring $GrCRing$ mapping $\mathcal{O}_X \mapsto \pi^*(\mathcal{O}_X)$, whose trivialization to the ordinary scheme is $sComm \to CRing$ mapping $\mathcal{O}_X(U) \mapsto \pi_0(\mathcal{O}_X(U))$.

[2] prove it



The truncation of derived scheme t_0 is taking π_0 on $(X, \mathcal{O}_X)$, denoted by $t_0 : dSch \to Sch$, and its inverse is full embedding $i^* : Sch \to dSch$ by using $i : CRing \to sComm$.

$$dSch \underset{i^*}{\overset{t_0}{\rightleftarrows}} Sch$$

One of natural question might be if the scheme $(X, t_0(\mathcal{O}_X))$ can reconstruct the original derived scheme? The existence of higher homotopy by Postnikov tower

$$\mathcal{O}_X \to \cdots \to t_{\leqq n}(\mathcal{O}_X) \to t_{\leqq n-1}(\mathcal{O}_X) \to \cdots \to t_0(\mathcal{O}_X) = \pi_0(\mathcal{O}_X)$$

making a sequence of derived schemes $\{(X, t_{\leqq n}(\mathcal{O}_X))\}$.

We define affine derived foliation using the definition of derive scheme. As noted later, affine derived foliation is a graded mixed cdga, whose category can be identified with derived stack. Intuitively, if foliation is a transformation of a manifold to a set of disjoint union, transformating derived scheme to derived foliation is to transform cdga to graded mixed cdga, which is many cdga.

Definition 1.3. Affine Derived Foliation

An affine derived foliation is a graded mixed cdga A with connectivity, perfectness, and quasi-freeness.

- (Connectivity)
  The underlying cdga $A(0)$ is cohomologically concentrated in non-positive degrees and is finitely presented over $\mathbb{C}$.
- (Perfectness)
  The $A(0)$-dg-module $A(1)[-1]$ is perfect and connective.
- (Quasi-freeness)
  The natural morphism of graded cdga's

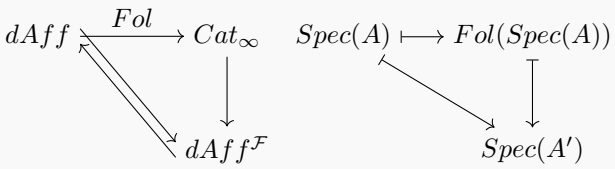
  $$Sym_{A(0)}(A(1)) \to A$$

  is a quasi-isomorphism of graded cdga.

Affine derived foliation generates a category $dAff^{\mathcal{F}}$, and it has a canonical ∞-functor $dAff^{\mathcal{F}} \to dAff$ with canonical mapping $A \mapsto Spec(A(0))$.

For the further formalization, an ∞-functor $Fol : dAff^{op} \to Cat_\infty$ desribes all the possible foliations, while $dAff^{\mathcal{F}}$ only describes one of them, namely a foliation is an object $\mathcal{F} \in Fol(X)$.

Definition 1.4. Derived Foliation



where $A = A'(0)$.

For the generalization of derived affine scheme $Spec(A)$ to derived stack $X \in dSt$, consider scheme is defined to be locally isomorphic to an affine scheme, but categorically reformulated to a projective limit of affine schemes. Accordingly, the foliation will be rewitten as

$$Fol(X) = \varprojlim_{Spec(A) \to \mathcal{F}} Fol(Spec(A))$$

where for definining derived stack dSt, we'll start from classical stack, and roughly speaking, stack is a 2-sheaf, in other words a functor from category of vector bundles to some category with Grothendieck topology and descent. Typically the codomain would be ∞-groupoid. For derived argument, derived prestack is category $P(dAff_k^{op}) = Fun(dAff_k^{op}, S)$, and derived stack is an embedding $Shv(dAff_k, \tau) \hookrightarrow Fun(dAff_k^{op}, S)$, this embedding functor has an adjoint, called stackification. This topology τ defines an accessible left-exact localization, this is a presentable ∞-category, uses as a replacement of Grothendieck topology.

Definition 1.5. Derived Stack

Definition 1.6. Lie Algebroid

Definition 1.7. Chevalley-Eilenberg Algebra

For a Lie algeboid T, CE algebra $CE^*(T)$ is.

$C^*(T) = Sym_{\mathcal{O}_X}(T^\vee[-1])$

CE algebra $CE^*(T)$ is used to compute derived foliation.

Definition 1.8

Definition 1.9. Perfect Crystal

Crystal is a vector bundle with flat connection. Perfect crystal is a crystal whose underlying complex is perfect.

Question 1.1

Is there any extension of foliation to Lagrangian foliation? To answer this question, all the previous foliation is complex dimension.

Definition 1.10

Definition 1.11

Definition 1.12

2 Structure of Quotient Stack

Definition 2.1

Quotient stack?

quotient?
Consider grouopid action functorially. $Fun(BG, X)$
Quotient stack has a

Example 2.1. Equivariant Intersection – Group

Consider a mapping $X \to [X/G]$ where G is a group.

Example 2.2. Equivariant Intersection – Groupoid

Definition 2.2

Definition 2.3

3 Groupoid Foliation

Definition 3.1

Definition 3.2

4 Quotient Stack

GIT?

5 Lagrangian Intersection Theory

The question is if there is any intersection theory of two foliations of Lagrangians.

Definition 5.1. Lagrangian

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